EVENT FORCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted in partial fulfilment of the requirements for the project component of the academic curriculum

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SALESFORCE DEVELOPER PROJECTS TRACK

Project Documentation

EVENT FORCE MANAGEMENT SYSTEM

Brief Contents

Introduction • Objectives • Requirements • Architecture • Data Model • Security • Event Module • Staff Module • Tasks Allocation • Dashboard Reports • Testing • Conclusion

ABSTRACT

Event Force Management System

The Event Force Management System is a software application designed to manage and organize the workforce required for various events, such as college functions, cultural programs, seminars, conferences, weddings, and public gatherings. The main objective of this project is to simplify the process of managing event staff, assigning duties, monitoring attendance, and coordinating event activities efficiently.

The proposed system helps the administrator maintain staff details, event information, duty assignments, and attendance records in a centralized database. It enables the administrator to allocate staff members to different events based on their availability and assigned responsibilities. The system also supports task management and report generation, helping improve coordination and reduce manual paperwork.

> ### Key System Purpose
>
> To overcome traditional workforce management difficulties, including duplicate records, scheduling conflicts, communication delays, and human errors in staff records through a centralized computerized database.

The Event Force Management System provides a simple, user-friendly, and organized approach to event workforce administration. In the future, the system can be enhanced with features such as automated duty scheduling, mobile notifications, QR-code-based attendance, and real-time event updates.

Overall, this project provides an efficient solution for managing event staff and coordinating workforce activities, ensuring smoother planning and execution of events across academic and enterprise institutions.

CHAPTER 1. INTRODUCTION

1.1 Introduction to Event Force Management System

The Event Force Management System is a software application developed to manage and coordinate the workforce involved in organizing different types of events. Events such as college functions, cultural programs, seminars, conferences, weddings, and public gatherings require proper planning, staff allocation, task assignment, and attendance monitoring. Managing these activities manually can be difficult, time-consuming, and prone to errors.

The main purpose of this system is to simplify event workforce management by providing a centralized platform for maintaining staff details, event information, duty schedules, and attendance records. The administrator can add and update event details, register staff members, assign duties, monitor attendance, and generate reports whenever required.

1.2 Project Overview

The Event Force Management System is designed to replace traditional manual methods of managing event staff with a computerized solution. It maintains organized records of events, employees or volunteers, assigned responsibilities, and attendance details.

The administrator can manage multiple events and allocate available staff members according to the requirements of each event. The system also helps track assigned tasks and maintain records for future reference.

1.3 Objectives of the Project

The main objectives of the Event Force Management System are:

- To maintain staff and volunteer information in a centralized database.
- To manage event details, including event names, dates, venues, and descriptions.
- To assign staff members to specific events and duties.
- To record and monitor staff attendance accurately.
- To manage tasks and responsibilities effectively.
- To reduce manual paperwork and administrative errors.
- To generate comprehensive reports related to events, staff, and attendance.
- To improve coordination and productivity during event planning and execution.

1.4 Scope of the Project

The scope of the Event Force Management System includes managing event information, staff registration, workforce allocation, attendance tracking, task assignment, and report generation.

The system can be used by event administrators to organize workforce activities and maintain accurate records. It provides a structured way to manage staff availability and responsibilities for different events.

In future versions, the system can be enhanced with mobile applications, automated scheduling, QR-code attendance, email or SMS notifications, and real-time updates.

CHAPTER 2: SYSTEM ANALYSIS

2.1 Existing System

The existing system for event force management generally depends on manual methods to organize events, maintain staff records, allocate duties, and track attendance. Event organizers often use paper records, registers, spreadsheets, and telephone communication to coordinate event activities.

In the manual system, the administrator records event details such as the event name, date, venue, and staff requirements. Staff information is maintained separately, and duties are assigned according to the requirements of each event.

Activities in the Existing System

- Event details are recorded manually on paper files.
- Staff information is maintained in physical registers or Excel spreadsheets.
- Staff duties are assigned manually based on informal memory or notes.
- Attendance is recorded using physical sign-in paper sheets.
- Task progress is monitored through manual follow-up calls.
- Reports are prepared by manually collecting and aggregating information from various physical logs.

2.2 Disadvantages of Existing System

The manual event management system has several key disadvantages that hinder operational efficiency:

1. **Time-Consuming Process:** Recording event details, assigning staff, and preparing reports manually consume significant administrative time.
2. **Data Duplication:** The same event or staff information is often recorded in multiple spreadsheets, leading to duplicate and conflicting data.
3. **Human Errors:** Manual data entry frequently causes mistakes in dates, contact details, and duty allocations.
4. **Difficult Record Maintenance:** Managing physical paper files becomes harder as the number of events increases.
5. **Poor Communication:** Last-minute updates to event schedules or duties fail to reach all staff members in time.
6. **Scheduling Conflicts:** Assigning staff without real-time tracking leads to overlapping shift allocations.
7. **Delayed Report Generation:** Manual report aggregation slows down decision-making.
8. **Risk of Data Loss:** Paper logs are vulnerable to damage, loss, or unauthorized tampering.

2.3 Proposed System

The proposed Event Force Management System is a web-based computerized application designed to streamline event workforce management. All data is securely stored in a centralized database accessible by administrators.

Key Features

- **Admin Authentication:** Secure password-protected login panel.
- **Event Management:** Full CRUD operations for creating and updating event schedules.
- **Staff Registration:** Structured management of staff contact details and default roles.
- **Dynamic Workforce Allocation:** Real-time mapping of available staff to events.
- **Task Tracking:** Clear duty assignments and progress tracking.
- **Automated Reporting:** Single-click generation of shift rosters and event summaries.

2.4 Advantages of Proposed System

- Eliminates paper registers and manual filing.
- Instant retrieval of event and staff information.
- Reduces clerical errors with field data validation.
- Centralized database ensures single source of truth.

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Hardware Requirements

Hardware requirements specify the minimum equipment needed to run the system smoothly:

Hardware Component	Minimum Specification
Processor	Intel Core i3 / AMD Ryzen 3 or equivalent
System Memory (RAM)	4 GB DDR4
Storage Space	10 GB available SSD/HDD storage
Display Device	1024x768 screen resolution standard monitor
Peripherals	Standard Keyboard and Mouse
Network Connectivity	Local Network / Internet Interface Card

These specifications ensure seamless responsiveness during concurrent admin operations and reporting tasks.

3.2 Software Requirements

Software Tool	Specification / Version
Operating System	Windows 10 / 11 or Linux (Ubuntu)
Front-End Interface	HTML5, CSS3, JavaScript (ES6)
Server Scripting	PHP 8.x
Database Engine	MySQL Server 8.0
Web Server Stack	Apache HTTP Server / XAMPP Stack
Development IDE	Visual Studio Code
Browser Support	Google Chrome / Microsoft Edge

3.3 Functional Requirements

- **Admin Authentication:** Verifies user credentials before opening admin dashboard.
- **Event Management:** Adds, modifies, or archives event records.
- **Staff Allocation:** Assigns personnel according to shift schedules and skill sets.
- **Report Generation:** Exports structured duty rosters and attendance summary reports.

3.4 Non-Functional Requirements

Non-functional requirements describe quality, performance, and operational constraints:

- **Performance:** System responds to database queries within less than 2 seconds.
- **Security:** Password fields are encrypted; unauthorized page access is blocked via session handling.
- **Usability:** Simple, intuitive interface requiring minimal training for users.
- **Reliability:** High operational availability with regular database transaction logging.
- **Maintainability:** Modular code separation between presentation, logic, and database layers.
- **Scalability:** Capable of supporting additional event records and staff expansion without performance degradation.
- **Data Integrity:** Foreign key constraints enforce relationship validity between events and staff records.

CHAPTER 4: SYSTEM DESIGN

4.1 System Architecture

The Event Force Management System employs a robust 3-Tier Software Architecture:

1. Presentation Layer (Front-End)

Built using HTML5, CSS3, and JavaScript to deliver responsive admin forms and interactive tables.

2. Application Layer (Business Logic)

Executed via PHP scripts to process form inputs, enforce shift rules, and manage admin session authentication.

3. Database Layer (Data Persistence)

Managed by MySQL relational database storing admin records, staff info, events, allocations, and tasks.

4.2 Data Flow Diagram (DFD)

The DFD models how data flows into, through, and out of the Event Force Management System.

Level 0 (Context Diagram)

The Administrator provides login credentials, event creation details, staff registrations, and allocation commands to the central system. In return, the system provides authentication status, dashboard views, roster reports, and confirmation notifications.

Level 1 Data Flow Processes

1. **1.0 User Authentication:** Validates admin credentials against stored records.
2. **2.0 Event Processing:** Manages event creation, venue allocation, and date updates.
3. **3.0 Staff Data Processing:** Registers and updates personnel background information.
4. **4.0 Roster & Duty Allocation:** Connects available staff to specific event shifts.
5. **5.0 Reporting Engine:** Compiles duty lists and attendance status into printable layouts.

4.3 Use Case Diagram Analysis

The Use Case Diagram defines the interactions between the primary actor (Administrator) and system modules.

> **Primary Actor: Administrator**
>
> Has complete administrative privileges across all system operations.

Key Use Cases

- **UC-1 Log In / Log Out:** Secure authentication and session teardown.
- **UC-2 Manage Events:** Create, edit, view, or remove event listings.
- **UC-3 Manage Staff Profiles:** Register personnel, edit contact numbers and roles.
- **UC-4 Allocate Staff:** Assign personnel to specific event duty rosters.
- **UC-5 Assign Tasks:** Detail specific operational responsibilities.
- **UC-6 Generate Reports:** Filter and print workforce reports.

4.4 Entity Relationship Diagram (ERD)

The ER Diagram maps entity attributes and structural relationships across database entities:

Entity Definitions

- **ADMIN:** Stores administrator security accounts (`admin_id`, `username`, `password_hash`).
- **EVENT:** Stores event operational data (`event_id`, `event_name`, `event_date`, `venue`).
- **STAFF:** Stores personnel background info (`staff_id`, `staff_name`, `phone`, `email`, `role`).
- **ALLOCATION:** Associative entity mapping staff to events (`allocation_id`, `event_id`, `staff_id`, `assigned_role`).
- **TASK:** Tracks specific duties attached to allocations (`task_id`, `allocation_id`, `task_description`, `status`).

Cardinals

One Event can have multiple Staff Allocations (1:N). One Staff member can participate in multiple Allocations over time (1:N). Each Allocation can break down into multiple specific Tasks (1:N).

4.5 Database Design Schema

Detailed relational structure for core database tables:

1. Admin Table (`admin`)

Field	Data Type	Constraints	Description
admin_id	INT	Primary Key, Auto Inc.	Unique Admin ID
username	VARCHAR(50)	NOT NULL, Unique	Login Username
password_hash	VARCHAR(255)	NOT NULL	Bcrypt Encrypted Hash

2. Event Table (`events`)

Field	Data Type	Constraints	Description
event_id	INT	Primary Key, Auto Inc.	Unique Event Identifier
event_name	VARCHAR(100)	NOT NULL	Title of Event
event_date	DATE	NOT NULL	Scheduled Date
venue	VARCHAR(150)	NOT NULL	Event Location
description	TEXT	NULL	Detailed Agenda
status	VARCHAR(30)	DEFAULT 'Planned'	Status Flag

3. Staff Table (`staff`)

Field	Data Type	Constraints	Description
staff_id	INT	Primary Key, Auto Inc.	Unique Personnel ID
staff_name	VARCHAR(100)	NOT NULL	Full Name
phone	VARCHAR(20)	NOT NULL	Mobile Number
email	VARCHAR(100)	NULL	Email Address
role	VARCHAR(50)	NOT NULL	Default Skill Role

4. Allocation Table (`allocations`)

Field	Data Type	Constraints	Description
allocation_id	INT	Primary Key, Auto Inc.	Unique Allocation ID
event_id	INT	Foreign Key (events)	Associated Event
staff_id	INT	Foreign Key (staff)	Assigned Staff
assigned_role	VARCHAR(50)	NOT NULL	Duty Role

5. Task Table (`tasks`)

Field	Data Type	Constraints	Description
task_id	INT	Primary Key, Auto Inc.	Task Unique ID
allocation_id	INT	Foreign Key (allocations)	Mapped Allocation
task_description	TEXT	NOT NULL	Specific Task Details
status	VARCHAR(30)	DEFAULT 'Pending'	Progress Status

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 Module Overview

The system is developed through six cohesive modules designed to fulfill core operational functions:

- Admin Login Module
- Event Management Module
- Staff Registration & Management Module
- Workforce Allocation Module
- Task Management Module
- Report Generation Module

5.2 Admin Login Module

Provides a secure gateway requiring username and password verification. Upon successful verification, an encrypted PHP session variable is registered to authorize navigation across administrative dashboards.

5.3 Event Management Module

Allows administrators to perform full lifecycle management of events. Admins can schedule upcoming dates, set venues, specify staffing quotas, and archive past events.

5.4 Staff Management Module

Handles registration of staff profiles and volunteers. Stores contact details, availability flags, and skill categories (e.g., Technical, Crowd Control, Registration Desk) to facilitate allocation.

5.5 Staff Allocation Module

Matches personnel to scheduled events. Admins select an active event, choose available staff members, and assign specific duty roles, generating clean relational records in the database.

5.6 Task Management Module

Breaks down event allocations into distinct actionable assignments. Allows organizers to monitor progress statuses (e.g., Pending, In Progress, Completed).

5.7 Report Generation Module

Queries database records to synthesize clean, formatted duty rosters, master staff directories, and event outcome reports for administrative review.

5.8 Implementation Integration Summary

All six modules seamlessly communicate: front-end interface inputs trigger server-side processing scripts, executing relational MySQL queries and returning dynamic visual feedback.

CHAPTER 6: SYSTEM TESTING

6.1 Testing Methodology Overview

System testing verifies that the software satisfies functional user requirements, maintains data integrity, and performs reliably under daily administrative operations.

6.2 Unit Testing

Unit testing checks individual software components independently before module integration:

- **Login Module Test:** Evaluated response to valid credentials vs invalid attempts.
- **Event Form Test:** Tested field validation on missing date or venue entries.
- **Staff Registration Test:** Verified email syntax and phone number length rules.
- **Database Insertion Test:** Ensured SQL insert functions properly create new rows.

6.3 Integration Testing

Integration testing verifies that separate modules communicate seamlessly when combined:

- Ensured that newly created events immediately populate the Staff Allocation dropdown menus.
- Verified that registering a staff member immediately updates allocation availability counts.
- Checked that updating an allocation status accurately reflects across generated summary reports.

6.4 System Testing

Validates the complete end-to-end operational flow of the Event Force Management System across diverse browsers and operating system environments, ensuring zero session leaks or broken navigation links.

6.5 Test Cases and Verification Results

TC ID	Test Scenario	Expected Result	Status
TC01	Login with valid admin credentials	Dashboard opens successfully	**PASS**
TC02	Login with incorrect password	Displays error alert message	**PASS**
TC03	Add new event details	Event record saved in database	**PASS**
TC04	Modify scheduled event venue	Database record updated	**PASS**
TC05	Register new staff member profile	Staff record created	**PASS**
TC06	Allocate staff member to event	Allocation record saved	**PASS**
TC07	Generate event workforce report	Formatted report displayed	**PASS**
TC08	Submit empty required input field	Validation error message shown	**PASS**
TC09	Session logout click	Session terminated safely	**PASS**

CHAPTER 7: OUTPUT AND RESULTS

7.1 Login Page Module Interface

The login page serves as the entry authentication portal for the system. It validates user inputs, enforces security, and redirects authorized administrators directly to the central dashboard.

> **Interface Components**
>
> - Username input field
> - Encrypted password field
> - Submit Login button
> - Dynamic error notification banner for incorrect attempts

All access attempts without valid active session keys are blocked and redirected back to this portal.

7.2 Admin Dashboard Interface

The central admin dashboard provides real-time oversight of system metrics and quick action shortcuts.

Key Metrics Widgets

- **Total Scheduled Events:** Live count of active events.
- **Registered Personnel:** Total staff count available in database.
- **Active Shift Allocations:** Current assignments in progress.
- **Upcoming Schedules:** Chronological calendar preview.

Intuitive side navigation allows instant access to Event Management, Staff Management, Allocation Tool, and Reporting Subsystems.

7.3 Event Details Management View

Displays detailed tables of scheduled events containing Event ID, Title, Scheduled Date, Venue, Staff Count Needed, and Status flags. Features interactive buttons for Editing or Archiving event records.

7.4 Staff Allocation Interface View

Allows admins to select an active event from a dynamic dropdown list, select available personnel, assign specific duty roles (e.g., Technical Support, Registration Coordinator, Security Lead), and commit the roster to database storage.

7.5 Reports and Output Summaries

The reporting engine yields clean, printable rosters summarizing event shifts, personnel duty listings, and attendance logs, improving accountability across operations.

CHAPTER 8: CONCLUSION AND FUTURE ENHANCEMENTS

8.1 Conclusion

The Event Force Management System provides a structured, automated solution replacing traditional manual record keeping. By centralizing staff info, shift allocations, and reporting into a digital platform, event coordination productivity is significantly improved.

8.2 System Limitations

- Requires stable network connectivity for web server access.
- Initial staff profiles and event details require manual setup.
- Standard version lacks automatic SMS notification integration.

8.3 Future Enhancements

- **Mobile App:** Native Android/iOS mobile application for staff shift views.
- **Automated Alerts:** Email and SMS notification dispatch upon shift assignment.
- **QR Code Attendance:** Contactless attendance logging via QR codes.
- **Smart Conflict Detection:** Algorithmic prevention of double-booking staff across overlapping schedules.

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APPENDIX A: SAMPLE SOURCE CODE

A.1 Admin Login Page Interface (`index.html`)

```
 <!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Event Force Management System - Admin Login</title>
  <style>
    body { font-family: Arial, sans-serif; background: #f1f5f9; }
    .login-card { width: 350px; margin: 100px auto; padding: 25px; background: #fff;
border-radius: 8px; }
    .btn-login { background: #1e40af; color: #fff; padding: 10px; border: none; width:
100%; cursor: pointer; }
  </style>
</head>
<body>
  <div class="login-card">
    <h2 style="color: #1e40af;">Admin Login</h2>
    <form action="login.php" method="POST">
      <label>Username:</label><br>
      <input type="text" name="username" required style="width: 100%; margin-bottom:
15px;"><br>
      <label>Password:</label><br>
      <input type="password" name="password" required style="width: 100%; margin-bottom:
20px;"><br>
      <button type="submit" class="btn-login">Authenticate</button>
    </form>
  </div>
</body>
</html>
```

A.2 Database Connection Handler (`db_config.php`)

```
 <?php
/* Database Configuration Parameters */
$servername = "localhost";
$username   = "root";
$password   = "";
$dbname     = "event_force_db";

// Establish Relational Database Connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Verify Connection Integrity
if ($conn->connect_error) {
    die("Database Connection Failed: " . $conn->connect_error);
}

// Set Character Set Encoding
$conn->set_charset("utf8mb4");
?>
```

This backend script establishes persistent server connections to MySQL, supporting parameterized query executions across all administrative workforce modules.

PROJECT COMPLETION & SIGN-OFF

This report documents the full software engineering lifecycle of the Event Force Management System developed for academic verification and deployment suitability.

Document Verification Summary

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Student Signature **Head of Department**